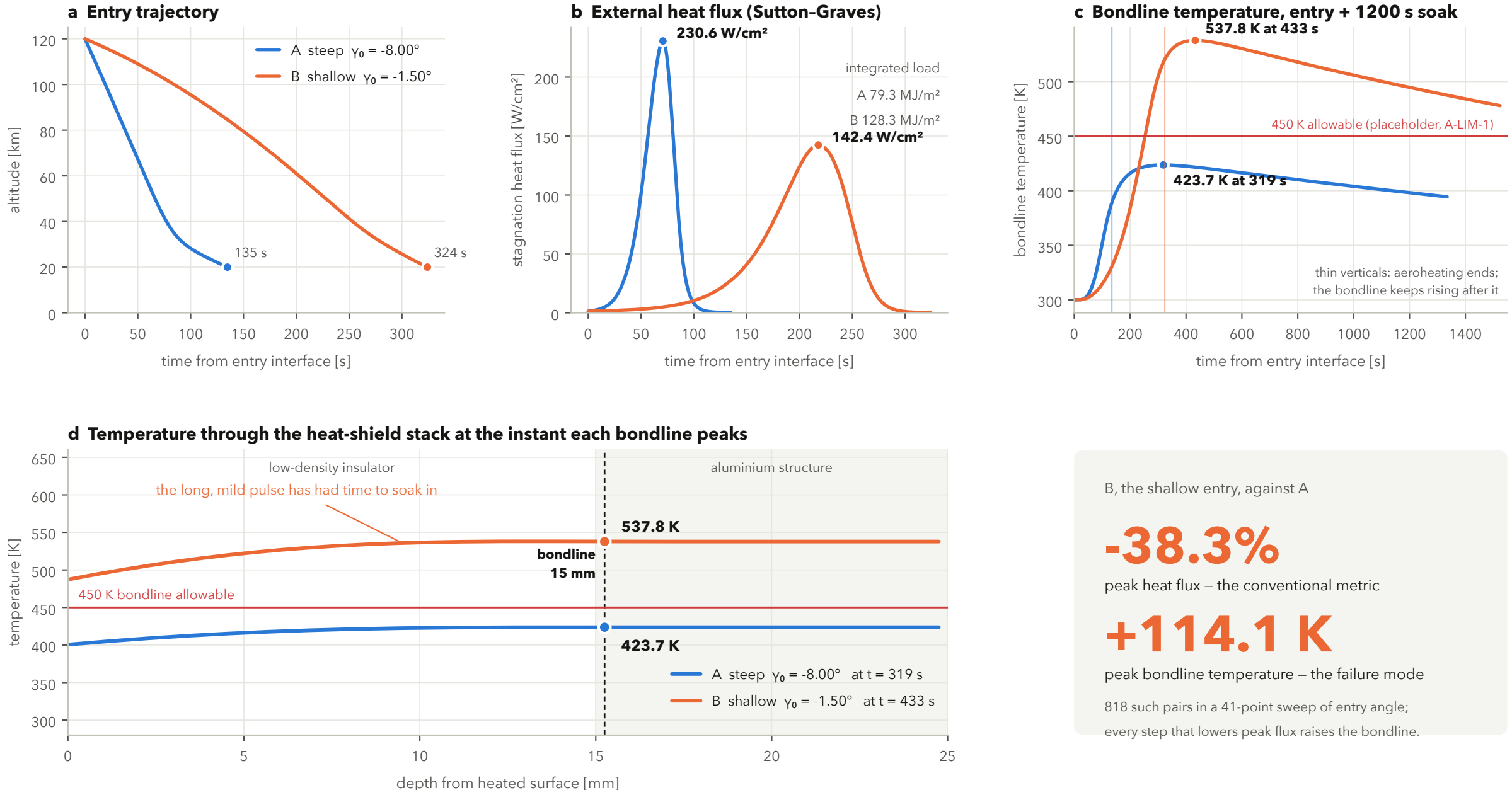


1

Burn versus bake: the shallow entry is cooler outside and hotter inside

Two entries of the same capsule. The shallow one lowers peak heat flux by 38.3% and leaves the bondline 114.1 K hotter. Peak flux is a surface quantity; bondline temperature is an integrated, delayed one.



B, the shallow entry, against A

-38.3%

peak heat flux – the conventional metric

+114.1 K

peak bondline temperature – the failure mode

818 such pairs in a 41-point sweep of entry angle;
every step that lowers peak flux raises the bondline.

Run M1-20260902T130547Z · git d841bf7 (working tree dirty) · config hash 056afae32fd1 · designs M1-g-08.00 (A) and M1-g-01.50 (B) · scalars as logged in results/M1/M1-20260902T130547Z/candidates.csv.

Fidelity 0 with the LEGACY cap-radius nose model ($R_{\text{eff}} = R_n$, harmless for this hemispherical nose), constant $C_D = 1.2$, Sutton-Graves convective stagnation heating, 1-D conduction, adiabatic back face, placeholder TPS properties.

Time histories were not persisted by the M1 run: they are re-derived here by evaluate_design() from config_snapshot.yaml and checked against the logged scalars to 1e-9 relative. Model predicts, within tested assumptions; no claim about any flight vehicle.